We've only scratched the surface of GMP here; the official PHP documentation for this extension is a good place to start for further info.

https://www.php.net/manual/en/book.gmp.php

There are also extensions for statistics (based on DCDFLIB and RANDLIB) in case you're in need of a cheeky chi-square or t-test, and financial trading (based on TA-Lib) if phrases such as "three-line strike" or "time series forecast" tend to crop up at your day job. These are quite specialized, as you'd expect, and beyond the scope of this text. Again, your starting point is the official PHP documentation.

https://www.php.net/manual/en/book.stats.php
https://www.php.net/manual/en/book.trader.php

Type Coercion

In Chapter 2, "Data, Types, and Variables," we saw how strings could be converted to numbers and vice versa. Previous versions of PHP had some problems with numeric and numerical strings (numeric means "all characters are digits," numerical means "some characters are digits") that could lead to security problems.

Explicitly Casting Strings to Numbers

Casting a numeric string (containing only valid number digits) to a number is straightforward; just remember that the only base support is Base 10. Numerical strings (containing non-digits or partially invalid digits) are a little trickier:

```
var_dump((int) 'speed: 100.1 mph'); // alphabet characters
                                    encountered first
var_dump((int) '100.1 mph'); // decimal point ignored
var_dump((float) "\t 100.1 mph"); // decimal point included
var_dump((float) '1,000,000'); // decimal 1000000 in common format
var_dump((int) '1f'); // hexadecimal 31
var_dump((int) 'ff'); // hexadecimal 255
var_dump((int) '0xff'); // same but using common hex notation
var_dump((int) '1111'); // binary 16
var_dump((int) '012'); // octal 10
```